Panskura Banamali College (Autonomous)

B.Sc.(H) 1st Semester Examination 2022 (Online mode)

Subject : Computer Science Paper: CC2P

Paper Name : Computer System Architecture Lab

F.M. : 20 Time : 2 Hours

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2021UG009967, paper_code : CC1P , then file name will be :: 2021UG009967- CC1P.pdf for question 'A' and 2021UG009967- CC1P.mp3 for question 'B']

A. Answer any one question: 1 x12 = 12

1. Design an Ex-OR gate using NOR gates only. Measure the input & output voltages. Establish the Truth Table.
2. Design a Mod-10 counter using T-F/F .Verify the result & draw the timing diagram.
   a) Design a of counter
   b) Verify of result
   c) Drawing of timing diagram
3. Design a full adder using basic logic gates & verify its truth table.
4. Design a 4 bit shift register using F/F & basic gates study the following modes of operation.
   a) Serial-in-parallel out.
   b) Serial-in-serial out.
5. Construct a 4:1 MUX using NAND gates only.
   a) Draw the circuit & obtain the truth table.
   b) Verify the truth table.
6. Design a logic circuit of the following function & verify its truth table &performance.
   F(A,B,C)=AC+BC+A
7. Design a 2 to 4 decoder circuit & verify its truth table.
8. Design a clocked J-K F/F using NOR gates only & verify its truth table.

B. Answer any four questions: 4 x2 = 8

[ONLINE Viva-Voce : Record the audio of your answers as like .mp3 format]

1. What is Encoder & Decoder?
2. What is Sequential circuit?
3. What is Combinational circuit?
4. Define Master Slave Flip Flop?
5. Define BCD ?
6. What is ROM?
7. What is register?
8. What is use of basic gate?
9. What is IC?
10. What is counter ?

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